

Finite-Sample Reconstruction: Rates, Witnesses, and the Honest Bridge

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1 The Bias Bound and the Algebra Side

1.1 The bias bound

Let $(X, \mathcal{B}, \mu)$ be a probability space and $\mathfrak{m} \subseteq \mathcal{B}$ a sub- σ -algebra. Define the σ -algebra approximation error

$$\delta(\mathfrak{m}) := \sup_{S \in \mathcal{B}} \inf_{E \in \mathfrak{m}} \mu(S \triangle E). \quad (1)$$

This measures how well $\mathfrak{m}$ approximates $\mathcal{B}$: it is zero if and only if $\mathfrak{m} = \mathcal{B}$ modulo μ -null sets.

Lemma 1.1 (Bias bound). *For every $f \in L^\infty(\mu)$,*

$$\|f - \mathbb{E}[f \mid \mathfrak{m}]\|_{L^2(\mu)} \leq 2\|f\|_{L^\infty} \cdot \delta(\mathfrak{m})^{1/2}.$$

Proof. Step 1 (Indicators). Fix $S \in \mathcal{B}$. By definition of $\delta(\mathfrak{m})$, there exists $E \in \mathfrak{m}$ with $\mu(S \triangle E) \leq \delta(\mathfrak{m})$. Since $\mathbb{1}_E \in L^2(X, \mathfrak{m}, \mu)$ and $\mathbb{E}[\mathbb{1}_S \mid \mathfrak{m}]$ is the L^2 -best $\mathfrak{m}$ -measurable approximation to $\mathbb{1}_S$,

$$\|\mathbb{1}_S - \mathbb{E}[\mathbb{1}_S \mid \mathfrak{m}]\|_{L^2}^2 \leq \|\mathbb{1}_S - \mathbb{1}_E\|_{L^2}^2 = \mu(S \triangle E) \leq \delta(\mathfrak{m}).$$

Step 2 (Bounded f via layer cake). For $f \in L^\infty(\mu)$ with $\|f\|_{L^\infty} \leq M$, the layer cake representation gives $f = \int_{-M}^M \mathbb{1}_{S_t} dt$ where $S_t = \{x : f(x) > t\}$. Since $\mathbb{E}[\cdot \mid \mathfrak{m}]$ is linear and continuous in L^2 ,

$$f - \mathbb{E}[f \mid \mathfrak{m}] = \int_{-M}^M (\mathbb{1}_{S_t} - \mathbb{E}[\mathbb{1}_{S_t} \mid \mathfrak{m}]) dt.$$

Applying Minkowski's inequality for integrals and Step 1:

$$\|f - \mathbb{E}[f \mid \mathfrak{m}]\|_{L^2} \leq \int_{-M}^M \|\mathbb{1}_{S_t} - \mathbb{E}[\mathbb{1}_{S_t} \mid \mathfrak{m}]\|_{L^2} dt \leq 2M \cdot \delta(\mathfrak{m})^{1/2}. \quad \square$$

Remark 1.2. The constant 2 is not optimal but the exponent $\frac{1}{2}$ is sharp: taking $X = [0, 1]$, $\mathfrak{m} = \{\emptyset, X\}$, $f = \mathbb{1}_{[0, 1/2]}$ gives $\delta(\mathfrak{m}) = \frac{1}{2}$ and $\|f - \mathbb{E}[f \mid \mathfrak{m}]\|_{L^2} = \frac{1}{2} = \delta(\mathfrak{m})^{1/2}/\sqrt{2}$.

1.2 The three-term decomposition

We now specialise to the delay setting. Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system and $h \in L^\infty(\mu)$. Write $\mathcal{O}_h^{(L)} = \sigma(\Phi_h^{(L)})$ for the observable σ -algebra at lag L , where $\Phi_h^{(L)}(x) = (h(x), h(Tx), \dots, h(T^L x))$, and $\mathcal{A}_h^{(L)}$ for the class of degree- D polynomials in the delay coordinates $\Phi_h^{(L)}(x)$. Let $\hat{f}_n^{(L)}$ denote the empirical risk minimiser over $\mathcal{A}_h^{(L)}$.

Lemma 1.3 (Three-term decomposition). *For any $f \in L^\infty(\mu)$, any $L \geq 0$, and any $D \geq 1$:*

$$\|f - \hat{f}_n^{(L)}\|_{L^2(\mu)} \leq \underbrace{2\|f\|_{L^\infty} \cdot \delta(L)^{1/2}}_{\text{bias}} + \underbrace{\inf_{g \in \mathcal{A}_h^{(L)}} \|\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g\|_{L^2(\mu)}}_{\text{approx.}} + \underbrace{\|g_D^* - \hat{f}_n^{(L)}\|_{L^2(\mu)}}_{\text{statistical}}, \quad (2)$$

where $\delta(L) := \delta(\mathcal{O}_h^{(L)})$ as in (1) and $g_D^* = \operatorname{argmin}_{g \in \mathcal{A}_h^{(L)}} \|\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g\|_{L^2(\mu)}$.

Proof. Triangle inequality applied to the decomposition $f - \hat{f}_n^{(L)} = (f - \mathbb{E}[f \mid \mathcal{O}_h^{(L)}]) + (\mathbb{E}[f \mid \mathcal{O}_h^{(L)}] - g_D^*) + (g_D^* - \hat{f}_n^{(L)})$, with the first term bounded by Lemma 1.1. $\square$

Remark 1.4 (Scope). Lemma 1.1 holds for any sub- σ -algebra $\mathfrak{m}$; no dynamical structure is needed. The dynamical hypothesis enters only through the approximation and statistical terms, which require the image $\Phi_h^{(L)}(X)$ to be a manageable geometric object.

1.3 The observable reverse lower bound

The three-term bound controls the forward error. For the stopping rule to be honest, we also need a reverse bound: when $\delta(L)$ is large, the estimator $\hat{f}_n^{(L)}$ cannot approximate f well, regardless of n .

Lemma 1.5 (Observable reverse lower bound). *There exists a universal constant $c > 0$ such that for any $f \in L^\infty(\mu)$ with $\|f\|_{L^\infty} \geq \frac{1}{2}$ and any $L \geq 0$:*

$$\|\mathbb{1}_S - \hat{f}_n^{(L)}\|_{L^2(\mu_n)} \geq \frac{\delta(L)^{1/2}}{2} - C' \cdot n^{-s/(2s+d)}$$

with probability tending to 1 as $n \rightarrow \infty$, where $S \in \mathcal{B}$ is the set achieving the supremum in (1) at lag L and $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$ is the empirical measure.

Proof. *Step A (Population CE gap).* Let $S^* \in \mathcal{B}$ achieve the supremum in $\delta(L) = \sup_S \inf_{E \in \mathcal{O}_h^{(L)}} \mu(S \triangle E)$. For any $E \in \mathcal{O}_h^{(L)}$:

$$\|\mathbb{1}_{S^*} - \mathbb{E}[\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)}^2 \leq \|\mathbb{1}_{S^*} - \mathbb{1}_E\|_{L^2(\mu)}^2 = \mu(S^* \triangle E) \geq \delta(L) - \varepsilon$$

for any $\varepsilon > 0$. Taking $\varepsilon \rightarrow 0$: $\|\mathbb{1}_{S^*} - \mathbb{E}[\mathbb{1}_{S^*} \mid \mathcal{O}_h^{(L)}]\|_{L^2(\mu)} \geq \delta(L)^{1/2}$.

Since the conditional expectation is the L^2 -projection, $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu)} \geq \delta(L)^{1/2}$ for all $g \in \mathcal{A}_h^{(L)} \subseteq L^2(X, \mathcal{O}_h^{(L)}, \mu)$.

Step B (Empirical vs population norm). By Hoeffding's inequality and the bounded difference property of μ_n , $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu_n)} \geq \|\mathbb{1}_{S^*} - g\|_{L^2(\mu)} - O(n^{-1/2})$ with high probability, uniformly over $g \in \mathcal{A}_h^{(L)}$.

Step C (Empirical norm vs $\hat{f}_n^{(L)}$). Since $\hat{f}_n^{(L)}$ minimises empirical L^2 error over $\mathcal{A}_h^{(L)}$, and $\|\mathbb{1}_{S^*} - g\|_{L^2(\mu_n)} \geq \delta(L)^{1/2} - O(n^{-1/2})$ for all $g \in \mathcal{A}_h^{(L)}$, we have $\|\mathbb{1}_{S^*} - \hat{f}_n^{(L)}\|_{L^2(\mu_n)} \geq \frac{\delta(L)^{1/2}}{2} - C'n^{-s/(2s+d)}$ with high probability, absorbing the $O(n^{-1/2})$ into the statistical rate. $\square$

Remark 1.6 (Closed loop). Lemmas 1.3 and 1.5 together bound $\|\mathbb{1}_{S^*} - \hat{f}_n^{(L)}\|_{L^2(\mu_n)}$ from both sides. The upper bound (Lemma 1.3) says the error is at most $2\|f\|_{L^\infty} \cdot \delta(L)^{1/2} + C' \cdot n^{-s/(2s+d)}$. The lower bound (Lemma 1.5) says it is at least $\frac{\delta(L)^{1/2}}{2} - C' \cdot n^{-s/(2s+d)}$. The detectability threshold is the n at which the statistical floor $C'n^{-s/(2s+d)}$ drops below $\frac{\delta(L)^{1/2}}{2}$: reconstruction becomes detectable at sample size $n \gtrsim \delta(L)^{-(2s+d)/s}$.